

- 1 (a) gravitational force provides the centripetal force B1
 $GMm/r^2 = mr\omega^2$ (must be in terms of ω) B1
 $r^3\omega^2 = GM$ and GM is a constant B1 [3]
- (b) (i) 1. for Phobos, $\omega = 2\pi/(7.65 \times 3600)$ C1
 $= 2.28 \times 10^{-4} \text{ rad s}^{-1}$
 $(9.39 \times 10^6)^3 \times (2.28 \times 10^{-4})^2 = 6.67 \times 10^{-11} \times M$ C1
 $M = 6.46 \times 10^{23} \text{ kg}$ A1 [3]
2. $(9.39 \times 10^6)^3 \times (2.28 \times 10^{-4})^2 = (1.99 \times 10^7)^3 \times \omega^2$ C1
 $\omega = 7.30 \times 10^{-5} \text{ rad s}^{-1}$ C1
 $T = 2\pi/\omega = 2\pi/(7.30 \times 10^{-5})$
 $= 8.6 \times 10^4 \text{ s}$
 $= 23.6 \text{ hours}$ A1 [3]
- (ii) *either* almost 'geostationary'
or satellite would take a long time to cross the sky B1 [1]

- 2 (a) e.g. moving in random (rapid) motion of molecules/atoms/particles